

DevOps CI/CD Pipeline Lab

Project 1 — Record of Commands & Execution Evidence

Name	Karan Patel
Roll No.	2301410015
Course	B.Tech CSE (Cybersecurity)
Institution	K.R. Mangalam University, Gurugram

Environment: Ubuntu 24.04.4 LTS (WSL) | Stack: Git, Python 3.12 / Flask, Jenkins 2.568.3, OpenJDK 21

This document is a clean, organized record of the terminal commands executed and screenshots captured while setting up a local DevOps CI/CD pipeline — covering environment provisioning, application development and testing, Git/GitHub version control, Jenkins installation and pipeline configuration, and automated deployment with verification.

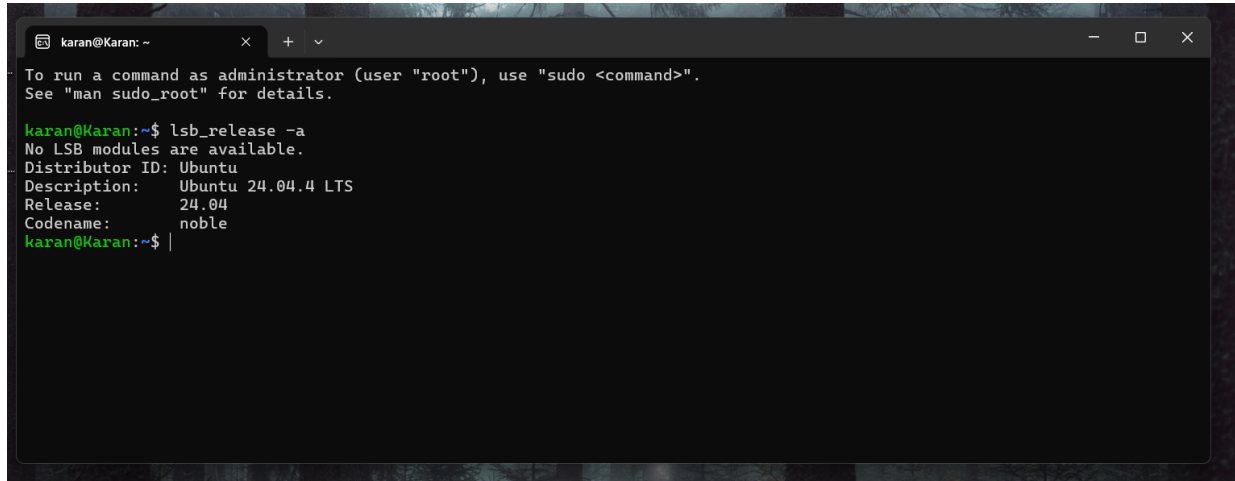
Contents

1. Environment & Tooling Setup
2. Jenkins Installation & Service Configuration
3. Application Development & Testing
4. Git & GitHub Version Control
5. Deployment & Verification

1. Environment & Tooling Setup

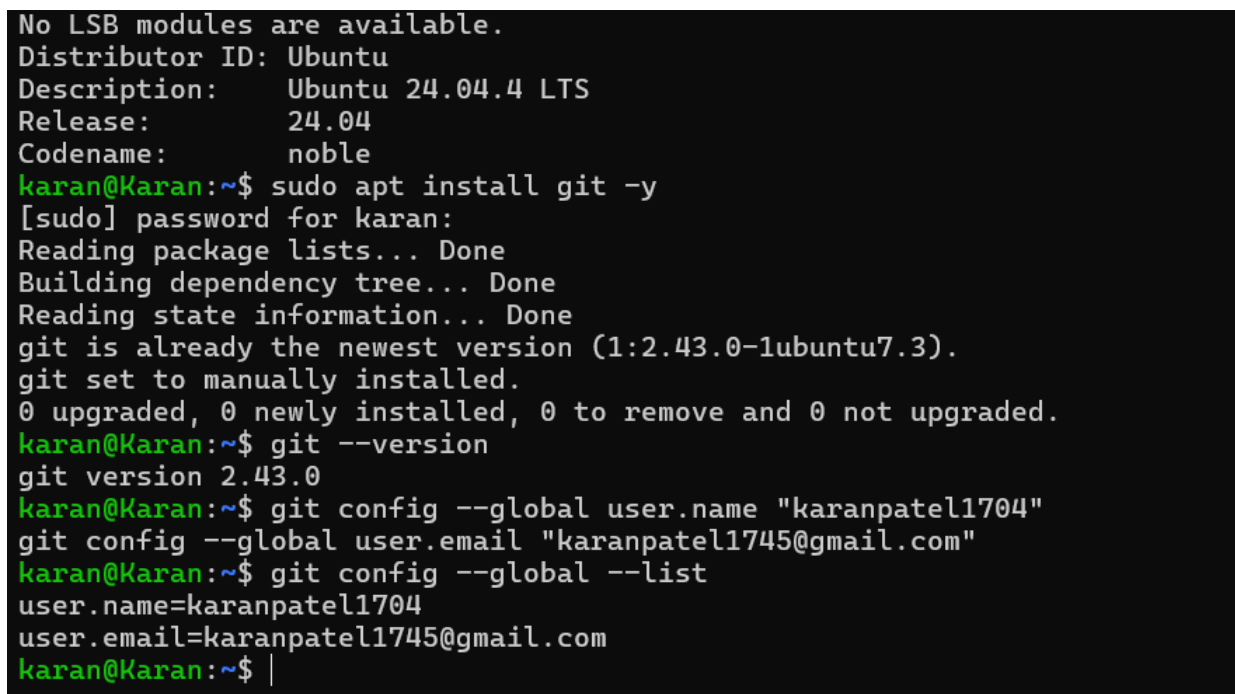
The lab began with confirming the OS environment and installing the core toolchain required for the pipeline: Git for version control, Python 3 for the sample application, and VS Code (via WSL) as the editor.

1.1 OS Verification & Git Installation



```
karan@Karan: ~  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
karan@Karan:~$ lsb_release -a  
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.4 LTS  
Release:        24.04  
Codename:       noble  
karan@Karan:~$ |
```

OS check (Ubuntu 24.04.4 LTS) and Git installation, followed by global Git identity configuration.



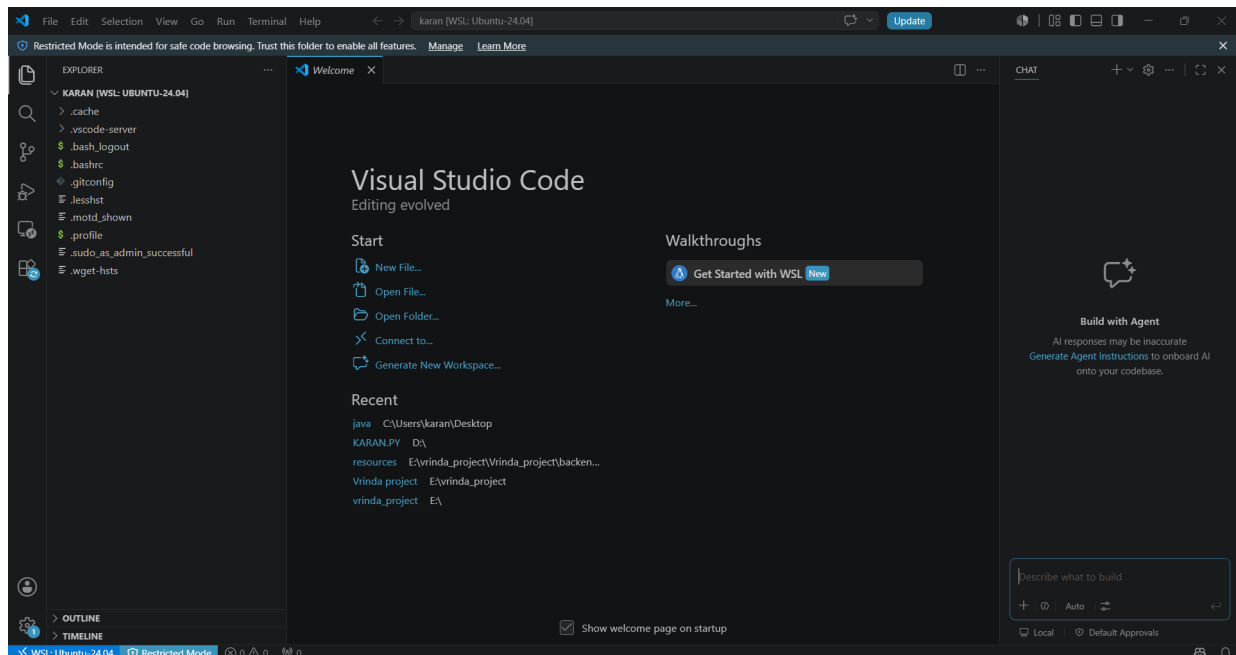
```
No LSB modules are available.  
Distributor ID: Ubuntu  
Description:    Ubuntu 24.04.4 LTS  
Release:        24.04  
Codename:       noble  
karan@Karan:~$ sudo apt install git -y  
[sudo] password for karan:  
Reading package lists... Done  
Building dependency tree... Done  
Reading state information... Done  
git is already the newest version (1:2.43.0-1ubuntu7.3).  
git set to manually installed.  
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.  
karan@Karan:~$ git --version  
git version 2.43.0  
karan@Karan:~$ git config --global user.name "karanpatel1704"  
git config --global user.email "karanpatel1745@gmail.com"  
karan@Karan:~$ git config --global --list  
user.name=karanpatel1704  
user.email=karanpatel1745@gmail.com  
karan@Karan:~$ |
```

Re-verification of OS release and Git installation with sudo authentication, confirming Git version 2.43.0.

1.2 Python Toolchain Installation

```
karan@Karan:~$ sudo add-apt-repository universe
Adding component(s) 'universe' to all repositories.
Press [ENTER] to continue or Ctrl-c to cancel.
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
karan@Karan:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
171 packages can be upgraded. Run 'apt list --upgradable' to see them.
karan@Karan:~$ sudo apt install python3 python3-pip python3-venv -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
python3 is already the newest version (3.12.3-0ubuntu2.1).
python3-pip is already the newest version (24.0+dfsg-1ubuntu1.3).
python3-venv is already the newest version (3.12.3-0ubuntu2.1).
0 upgraded, 0 newly installed, 0 to remove and 171 not upgraded.
karan@Karan:~$ python3 --version
python3.12.3
karan@Karan:~$ pip --version
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
karan@Karan:~$
```

Adding the 'universe' repository, updating package lists, and installing python3, python3-pip and python3-venv.



Build dependencies (python3.12-dev, g++, build-essential) configured; Python 3.12.3 and pip 24.0 confirmed.

1.3 Editor Setup

```
karan@Karan: ~  
Setting up python3.12-dev (3.12.3-1ubuntu0.16) ...  
Setting up g++-x86-64-linux-gnu (4:13.2.0-7ubuntu1) ...  
Setting up g++-13 (13.3.0-6ubuntu2~24.04.1) ...  
Setting up libpython3-dev:amd64 (3.12.3-0ubuntu2.1) ...  
Setting up python3-dev (3.12.3-0ubuntu2.1) ...  
Setting up g++ (4:13.2.0-7ubuntu1) ...  
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode  
Setting up build-essential (12.10ubuntu1) ...  
Setting up libheif1:amd64 (1.17.6-1ubuntu4.8) ...  
Setting up libgd3:amd64 (2.3.3-9ubuntu5) ...  
Setting up libc-devtools (2.39-0ubuntu8.8) ...  
Setting up libheif-plugin-aomdec:amd64 (1.17.6-1ubuntu4.8) ...  
Setting up libheif-plugin-aomenc:amd64 (1.17.6-1ubuntu4.8) ...  
Processing triggers for libc-bin (2.39-0ubuntu8.8) ...  
Processing triggers for systemd (255.4-1ubuntu8.12) ...  
Processing triggers for man-db (2.12.0-4build2) ...  
karan@Karan:~$ python3 --version  
Python 3.12.3  
pip3 --version  
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)  
karan@Karan:~$
```

VS Code opened against the WSL Ubuntu-24.04 environment for editing the project files.

2. Jenkins Installation & Service Configuration

Jenkins requires a Java runtime, so OpenJDK 21 was installed first. The Jenkins Debian repository was then added, Jenkins was installed, and the service was enabled and started via systemd.

2.1 Java Runtime & Jenkins Repository Setup

```
karan@Karan:~$ sudo apt update
sudo apt install fontconfig openjdk-21-jre -y
java -version
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:2 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
171 packages can be upgraded. Run 'apt list --upgradable' to see them.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
fontconfig is already the newest version (2.15.0-1.1ubuntu2).
openjdk-21-jre is already the newest version (21.0.12+8-1~24.04).
0 upgraded, 0 newly installed, 0 to remove and 171 not upgraded.
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-24.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-24.04-Ubuntu, mixed mode, sharing)
karan@Karan:~$
```

Package index updated and fontconfig / openjdk-21-jre installed from the Jenkins Debian repository; Java 21 confirmed.

```
karan@Karan:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-24.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-24.04-Ubuntu, mixed mode, sharing)
```

OpenJDK 21.0.12 runtime environment verified as the active Java version.

2.2 Jenkins Service Activation

```
karan@Karan:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
karan@Karan:~$ sudo systemctl start jenkins
karan@Karan:~$ sudo systemctl status jenkins
● jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; pres>
   Active: active (running) since Wed 2026-09-09 14:12:43 UTC; 3min 52s a>
   Main PID: 5811 (java)
     Tasks: 53 (limit: 9302)
    Memory: 544.8M (peak: 784.7M)
       CPU: 12.905s
    CGroup: /system.slice/jenkins.service
           └─5811 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/

Sep 09 14:12:38 Karan jenkins[5811]: [LF]> This may also be found at: /var/
Sep 09 14:12:38 Karan jenkins[5811]: [LF]>
Sep 09 14:12:38 Karan jenkins[5811]: [LF]> *****
Sep 09 14:12:38 Karan jenkins[5811]: [LF]> *****
Sep 09 14:12:38 Karan jenkins[5811]: [LF]> *****
Sep 09 14:12:43 Karan jenkins[5811]: 2026-09-09 14:12:43.906+0000 [id=71] >
Sep 09 14:12:43 Karan jenkins[5811]: 2026-09-09 14:12:43.921+0000 [id=40] >
Sep 09 14:12:43 Karan systemd[1]: Started jenkins.service - Jenkins Continu>
Sep 09 14:12:46 Karan jenkins[5811]: 2026-09-09 14:12:46.020+0000 [id=88] >
Sep 09 14:12:46 Karan jenkins[5811]: 2026-09-09 14:12:46.020+0000 [id=88] >
lines 1-20/20 (END)
```

Jenkins service enabled at boot via systemd, started, and confirmed 'active (running)' with its process (PID) details.

2.3 Jenkins Dashboard

The screenshot displays the Jenkins web dashboard interface. At the top, there's a navigation bar with the Jenkins logo and a search icon. Below the navigation bar, the main content area is titled 'Welcome to Jenkins!'. It includes a message stating 'This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.' There are two main sections: 'Start building your software project' with a 'Create a job' button, and 'Set up a distributed build' with buttons for 'Set up an agent', 'Configure a cloud', and 'Learn more about distributed builds'. On the left side, there's a sidebar with a 'Build Queue' section showing 'No builds in the queue' and a 'Build Executor Status' section showing '0/2' executors. The bottom right corner of the dashboard shows 'REST API' and 'Jenkins 2.568.3'.

Jenkins web dashboard reachable at <http://localhost:8080>, confirming a successful installation (Jenkins 2.568.3).

3. Application Development & Testing

A sample Flask application was scaffolded under `~/devops-lab` with dedicated app, tests, and deploy directories. A virtual environment was created, dependencies installed, and the app validated with `pytest` before being run manually to confirm it served requests correctly.

3.1 Project Scaffolding & Repository Initialization

```
./app/app.py
./command-log.txt
./requirements.txt
./tests/test_app.py
(.venv) karan@karan:~/devops-lab$ cd ~/devops-lab
(.venv) karan@karan:~/devops-lab$ pwd
/home/karan/devops-lab
(.venv) karan@karan:~/devops-lab$ nano .gitignore
(.venv) karan@karan:~/devops-lab$ git status
fatal: not a git repository (or any of the parent directories): .git
(.venv) karan@karan:~/devops-lab$ git init
git branch -M main
hint: Using 'master' as the name for the initial branch. This default branch
name
hint: is subject to change. To configure the initial branch name to use in a
ll
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command
:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/karan/devops-lab/.git/
(.venv) karan@karan:~/devops-lab$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        app/
        command-log.txt
        requirements.txt
        tests/

nothing added to commit but untracked files present (use "git add" to track)
(.venv) karan@karan:~/devops-lab$
```

Project files listed, .gitignore created, and the local Git repository initialized (git init, branch renamed to main).

3.2 Test Execution

```
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/karan/d
evops-lab/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/karan/devops-lab
collected 0 items / 1 error

===== ERRORS =====
----- ERROR collecting tests/test_app.py -----
ImportError while importing test module '/home/karan/devops-lab/tests/test_a
pp.py':
Hint: make sure your test modules/packages have valid Python names.
Traceback:
/usr/lib/python3.12/importlib/__init__.py:90: in import_module
    return _bootstrap._gcd_import(name[level:], package, level)
    ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
tests/test_app.py:1: in <module>
    from app.app import app
E   ModuleNotFoundError: No module named 'app'
===== short test summary info =====
ERROR tests/test_app.py
!!!!!!!!!!!!!!!! Interrupted: 1 error during collection !!!!!!!!!!!!!!!!!!!!!!!
===== 1 error in 0.06s =====
(.venv) karan@Karan:~/devops-lab$ ls -la app
total 12
drwxr-xr-x 2 karan karan 4096 Sep  9 14:34 .
drwxr-xr-x 8 karan karan 4096 Sep  9 14:35 ..
-rw-r--r-- 1 karan karan   0 Sep  9 14:34 __init__.py
-rw-r--r-- 1 karan karan 246 Sep  9 14:33 app.py
(.venv) karan@Karan:~/devops-lab$ PYTHONPATH=. pytest -v
===== test session starts =====
platform linux -- Python 3.12.3, pytest-9.1.1, pluggy-1.6.0 -- /home/karan/d
evops-lab/.venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/karan/devops-lab
collected 2 items

tests/test_app.py::test_home PASSED [ 50%]
tests/test_app.py::test_health PASSED [100%]

===== 2 passed in 0.07s =====
(.venv) karan@Karan:~/devops-lab$
```

Initial pytest run failed with `ModuleNotFoundError` due to import path; re-run with `PYTHONPATH=.` resolved it, and both tests (`test_home`, `test_health`) passed.

3.3 Local Application Run

```
-bash: /home/karan/devops-lab: Is a directory
(.venv) karan@Karan:~/devops-lab$ python app/app.py
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deploym
ent. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.29.135.56:5000
Press CTRL+C to quit
```

Flask development server started via `'python app/app.py'`, serving the app on 127.0.0.1:5000 and the WSL network interface.

4. Git & GitHub Version Control

With the application and tests in place, all project files were staged and committed, and the local repository was connected to a remote on GitHub for version control and as the source for the Jenkins pipeline.

4.1 Staging Project Files

```
nothing added to commit but untracked files present (use "git add" to track)
(.venv) karan@Karan:~/devops-lab$ git status
On branch main

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore
        app/
        command-log.txt
        requirements.txt
        tests/

nothing added to commit but untracked files present (use "git add" to track)
(.venv) karan@Karan:~/devops-lab$ git add .
(.venv) karan@Karan:~/devops-lab$ git status
On branch main

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   .gitignore
        new file:   app/__init__.py
        new file:   app/app.py
        new file:   command-log.txt
        new file:   requirements.txt
        new file:   tests/test_app.py
```

Untracked files identified with `git status`, then staged using `'git add .'` and re-verified as ready to commit.

4.2 Initial Commit

```
(.venv) karan@Karan:~/devops-lab$ git commit -m "Initial commit: add sample
application and tests"
[main (root-commit) b036d23] Initial commit: add sample application and test
s
6 files changed, 39 insertions(+)
create mode 100644 .gitignore
create mode 100644 app/__init__.py
create mode 100644 app/app.py
create mode 100644 command-log.txt
create mode 100644 requirements.txt
create mode 100644 tests/test_app.py
(.venv) karan@Karan:~/devops-lab$ git log --oneline
b036d23 (HEAD -> main) Initial commit: add sample application and tests
(.venv) karan@Karan:~/devops-lab$
```

Initial commit created (6 files, 39 insertions) and confirmed via `'git log --oneline'`.

4.3 Pushing to GitHub

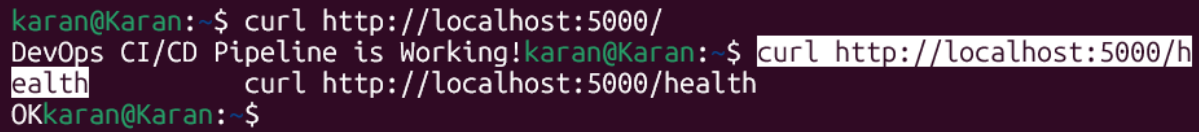
```
(.venv) karan@Karan:~/devops-lab$ git push -u origin main
Username for 'https://github.com': git status
Password for 'https://git%20status@github.com':
(.venv) karan@Karan:~/devops-lab$ git push -u origin main
Username for 'https://github.com': karanpatel1704
Password for 'https://karanpatel1704@github.com':
(.venv) karan@Karan:~/devops-lab$ git push -u origin main
Username for 'https://github.com': karanpatel1704
Password for 'https://karanpatel1704@github.com':
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 12 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (9/9), 919 bytes | 919.00 KiB/s, done.
Total 9 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/karanpatel1704/devops-sample-app1.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
(.venv) karan@Karan:~/devops-lab$ cd ~/devops-lab
git push -u origin main
Username for 'https://github.com':
```

Remote push authenticated with GitHub credentials; objects enumerated, compressed and pushed to 'karanpatel1704/devops-sample-app1', with the main branch tracking 'origin/main'.

5. Deployment & Verification

Following the Jenkins pipeline build (checkout → test → deploy stages defined in the Jenkinsfile) and the deployment script execution, the running application endpoints were verified directly with curl.

5.1 Endpoint Verification



```
karan@Karan:~$ curl http://localhost:5000/  
DevOps CI/CD Pipeline is Working!karan@Karan:~$ curl http://localhost:5000/h  
ealth      curl http://localhost:5000/health  
OKkaran@Karan:~$
```

Root endpoint returns 'DevOps CI/CD Pipeline is Working!' and the /health endpoint returns 'OK', confirming a successful end-to-end deployment.

End of record. This document consolidates the terminal command history and corresponding screenshots captured during the completed DevOps lab exercise.